

parts of the 3D model. Texture maps are essential to create a surface that mimics real life. Different Maps contain different information which makes up the whole surface in real life. The type of maps you use will be dependent on the asset & the art direction.

Texture types are the different types of maps used for different effects, which will depend upon the game engine and the visual style required.

1. Colour Maps

The most common colour map is diffuse. The diffuse map only represents base colours. This map varies in style and appearance depending on the requirements of the project. Diffuse maps are also used to create albedo map, which is similar to diffuse but without any shadows or highlights. Albedo is a more functional form of diffuse used by PBR shaders.

2. Transparency Maps

Transparency maps also called as opacity or alpha maps are used to cut out parts of a surface. Transparency is usually stored in what is known as the alpha channel of a diffuse map. These maps can define the parts of a model which would be visible.

3. Bump Maps

Bump maps are grayscale images which modify the shading properties of a mesh. These are used to create the illusion of fine details on a surface which would be hard to model. Bump maps simulate a 3D appearance, without it actually being modelled. For example, a simple cuboid can represent a wall, with the details of the 3D bricks showing up through a bump map.

4. Normal Maps

Normal maps have a similar function like bump maps. These are 3D bump maps which modify the vertex normals of a mesh to produce details on a mesh. Normal maps are commonly used to fake details on a low poly mesh.

5. Displacement Maps / Height Maps

Displacement maps are comparable to bump maps but store height information & modify the overall geometry when rendered. The difference between normal and displacement map is that displacement maps affect the silhouette of the model whereas normal maps only create the illusion of high details.